Course 4023 | Semester IV | Theory | 4 Credits

Industrial Engineering

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4023 as Industrial Engineering in Semester IV for Mechanical Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public industrial-engineering curriculum sources and is not represented as recovered official SITTTR Course 4023 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. The handbook uses transparent public-source themes—work measurement, productivity, quality, planning, layout and safety—and must be reconciled with restored official SITTTR details.

Verified source note: The official detailed source was inaccessible. Public sources used for work measurement, productivity, quality improvement, production planning, facilities and safety are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Industrial Engineering applies systematic methods to improve how people, materials, information and equipment work together. It combines work study, productivity, quality, production planning, facilities and safety to achieve efficient, safe and responsible operations.

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Prerequisite foundation

Basic arithmetic, engineering drawing/process awareness, data recording and an interest in systematic problem solving.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the scope, systems view and ethical responsibilities of industrial engineering.
2. Apply basic work-study and work-measurement methods to improve a process.
3. Use elementary productivity, quality and production-planning concepts for operational decisions.
4. Analyse facility flow, safety and continuous-improvement opportunities in a simple industrial case.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain industrial-engineering systems, productivity and improvement concepts.	Understand
CO2	Apply process charts, motion economy and time-study principles to a simple task.	Apply
CO3	Use basic quality, production-planning and inventory concepts to support operational decisions.	Apply
CO4	Analyse facility-flow, safety and improvement opportunities using data and stakeholder awareness.	Analyse

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Industrial Engineering and Productivity	Systems thinking, productivity, operations, resources, value, waste, costs, ergonomics and responsible improvement.	Approx. 14 hours	CO1	Module 1
2	Work Study and Work Measurement	Method study, process charts, motion economy, time study, work sampling, allowances and standard time.	Approx. 16 hours	CO2	Module 2
3	Quality, Planning and Control	Quality concepts, variation, basic quality tools, process capability awareness, forecasting, scheduling, inventory and project control.	Approx. 16 hours	CO3	Module 3
4	Facilities, Safety and Improvement Projects	Facility layout, material handling, line balancing, safety systems, risk awareness, project communication and improvement implementation.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Industrial Engineering and Productivity

Systems thinking, productivity, operations, resources, value, waste, costs, ergonomics and responsible improvement.

CO1 · Approx. 14 hours

Work Study and Work Measurement

Method study, process charts, motion economy, time study, work sampling, allowances and standard time.

CO2 · Approx. 16 hours

Quality, Planning and Control

Quality concepts, variation, basic quality tools, process capability awareness, forecasting, scheduling, inventory and project control.

CO3 · Approx. 16 hours

Facilities, Safety and Improvement Projects

Facility layout, material handling, line balancing, safety systems, risk awareness, project communication and improvement implementation.

CO4 · Approx. 14 hours

MODULE 1

Industrial Engineering and Productivity

Systems thinking, productivity, operations, resources, value, waste, costs, ergonomics and responsible improvement.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

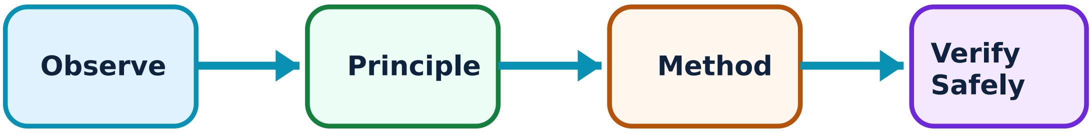
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Industrial Engineering and Productivity: identify the system, state its principle, follow the correct method and verify the result safely.

1. Industrial-engineering systems view−

Industrial engineering studies the interaction of people, materials, machines, methods, information and environment. Improvement seeks better safety, quality, delivery, cost and worker well-being rather than output alone.

Study and application method

Map a simple process from supplier/input through transformation to customer/output; identify measures for safety, quality, time, cost and service.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map a simple process from supplier/input through transformation to customer/output; identify measures for safety, quality, time, cost and service.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not optimise one local activity if it creates delay, risk or waste elsewhere in the system.

2. Productivity and performance measures−

Productivity relates output to inputs such as labour, material, energy or capital. Measures are meaningful only with a clear scope, time period, quality condition and ethical interpretation.

Study and application method

Define numerator, denominator, units, baseline and exclusions; compare like with like and explain likely causes before proposing action.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Higher output is not improved productivity if defects, injuries, overtime or resource use increase disproportionately.

3. Waste, value and continuous improvement−

Waste is effort or resource use that does not create required customer value. Continuous improvement uses observation, data, experimentation and standardisation to improve the process safely.

Study and application method

Observe the work, distinguish value-added from necessary/non-value-added steps, identify a small testable improvement and record the outcome.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Observe the work, distinguish value-added from necessary/non-value-added steps, identify a small testable improvement and record the outcome.

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Common mistake / precaution: Never implement a productivity change without considering safety, quality and worker input.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Work Study and Work Measurement

Method study, process charts, motion economy, time study, work sampling, allowances and standard time.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

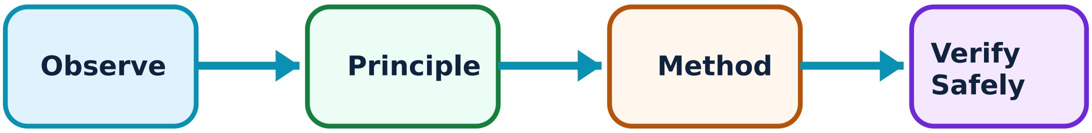
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Work Study and Work Measurement: identify the system, state its principle, follow the correct method and verify the result safely.

1. Method study and process charts−

Method study records and critically examines how work is performed to eliminate unnecessary movement, delays and handling. Operation, flow, two-hand and worker-machine charts make hidden process steps visible.

Study and application method

Define scope, observe actual work ethically, use consistent symbols, ask purpose/place/sequence/person/means questions and propose a safer simpler method.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define scope, observe actual work ethically, use consistent symbols, ask purpose/place/sequence/person/means questions and propose a safer simpler method.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not measure or judge workers without transparency, consent through proper workplace process and attention to fair conditions.

2. Motion economy and ergonomics−

Motion-economy principles reduce unnecessary reach, search, force and fatigue. Ergonomic design considers human capability, posture, repetition, lighting, tools and work height.

Study and application method

Identify high-frequency motions, layout/reach issues and awkward postures; propose changes that reduce strain while preserving safety and quality.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify high-frequency motions, layout/reach issues and awkward postures; propose changes that reduce strain while preserving safety and quality.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not use an average worker assumption to ignore accessibility, fatigue or injury risk.

3. Time study and standard time−

Time study observes defined task elements, rates normal performance and adds appropriate allowances to establish a fair standard time. Work sampling estimates proportion of time spent in states through random observations.

Study and application method

Break task into observable elements, collect enough representative observations, document rating/allowance assumptions and report uncertainty.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Break task into observable elements, collect enough representative observations, document rating/allowance assumptions and report uncertainty.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A standard time is not a target to pressure workers; it must be reviewed when method, conditions or product changes.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Quality, Planning and Control

Quality concepts, variation, basic quality tools, process capability awareness, forecasting, scheduling, inventory and project control.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

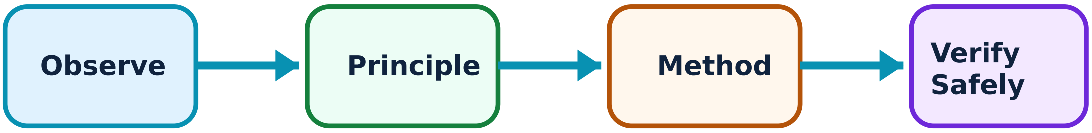
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Quality, Planning and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Quality and variation−

Quality means consistently meeting stated requirements. Variation may be common-cause (systemic) or special-cause (assignable); understanding the difference supports appropriate corrective action.

Study and application method

Define customer requirement and measurement method, collect data, plot/stratify it and use cause-and-effect or Pareto analysis before selecting action.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define customer requirement and measurement method, collect data, plot/stratify it and use cause-and-effect or Pareto analysis before selecting action.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not adjust a stable process for every small fluctuation; over-adjustment can increase variation.

2. Basic quality-improvement tools−

Check sheets, Pareto charts, cause-and-effect diagrams, flowcharts, histograms and control charts support structured problem solving and evidence-based decisions.

Study and application method

Choose the tool that fits the question: frequency, distribution, sequence, causes or stability; state data source and sampling limitations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Choose the tool that fits the question: frequency, distribution, sequence, causes or stability; state data source and sampling limitations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Charts do not replace investigation—correlation, ranking or a visible pattern is not proof of root cause.

3. Production planning and inventory−

Planning coordinates demand, capacity, materials, routing, scheduling and inventory. Inventory buffers uncertainty but creates holding, obsolescence and cash costs; schedules must respect real constraints.

Study and application method

List demand, lead time, capacity, batch/lot assumptions and constraints; prepare a simple sequence and identify the bottleneck/risk.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List demand, lead time, capacity, batch/lot assumptions and constraints; prepare a simple sequence and identify the bottleneck/risk.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not promise schedules without checking material availability, quality capacity, maintenance and human workload.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Facilities, Safety and Improvement Projects

Facility layout, material handling, line balancing, safety systems, risk awareness, project communication and improvement implementation.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

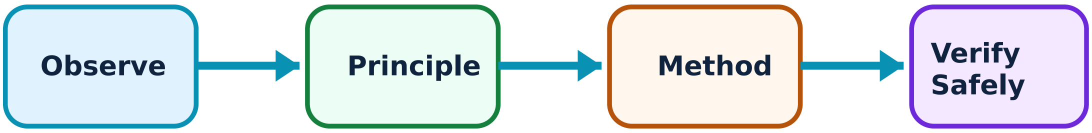
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Facilities, Safety and Improvement Projects: identify the system, state its principle, follow the correct method and verify the result safely.

1. Facility layout and material flow−

Facility layout positions resources to support flow, safety, supervision, flexibility and cost. Product, process, cellular and fixed-position layouts suit different volume/variety conditions.

Study and application method

Draw a spaghetti/flow diagram, quantify trips or distance where appropriate and compare layout alternatives against safety, throughput and flexibility criteria.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a spaghetti/flow diagram, quantify trips or distance where appropriate and compare layout alternatives against safety, throughput and flexibility criteria.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not reduce travel distance by creating unsafe crossings, congestion or inaccessible emergency routes.

2. Safety and risk control−

Industrial safety identifies hazards, assesses risk and applies controls using the hierarchy of elimination, substitution, engineering controls, administrative controls and PPE.

Study and application method

Describe task, hazard, exposure and consequence; choose the highest feasible control, assign responsibility and verify effectiveness.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe task, hazard, exposure and consequence; choose the highest feasible control, assign responsibility and verify effectiveness.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: PPE is important but is not the first or only control for a serious hazard.

3. Improvement project communication−

An improvement project needs a clear problem, baseline, target, evidence, tested change, implementation plan and follow-up measure. Stakeholder involvement improves feasibility and ownership.

Study and application method

Prepare a one-page A3-style report: background, current state/data, root causes, countermeasures, owner/timeline, result and next review.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a one-page A3-style report: background, current state/data, root causes, countermeasures, owner/timeline, result and next review.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not claim savings or quality gains without a defined baseline and a sustained post-change check.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Industrial Engineering and Productivity

Use the concept flow to revise observation, principle, method and verification.

Module 2: Work Study and Work Measurement

Use the concept flow to revise observation, principle, method and verification.

Module 3: Quality, Planning and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Facilities, Safety and Improvement Projects

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Map a simple workshop/service process and identify safety, quality and time metrics.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Conduct an instructor-approved method-study observation and prepare a process chart.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Use a check sheet and Pareto chart on a small simulated defect dataset.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a facility-flow and improvement proposal that includes risk controls and stakeholder feedback.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Industrial Engineering and Productivity

Explain Industrial-engineering systems view and state one correct method, application or precaution.−

Industrial engineering studies the interaction of people, materials, machines, methods, information and environment. Improvement seeks better safety, quality, delivery, cost and worker well-being rather than output alone.

Answer framework: Map a simple process from supplier/input through transformation to customer/output; identify measures for safety, quality, time, cost and service.

Important point: Do not optimise one local activity if it creates delay, risk or waste elsewhere in the system.

Explain Productivity and performance measures and state one correct method, application or precaution.−

Productivity relates output to inputs such as labour, material, energy or capital. Measures are meaningful only with a clear scope, time period, quality condition and ethical interpretation.

Answer framework: Define numerator, denominator, units, baseline and exclusions; compare like with like and explain likely causes before proposing action.

Important point: Higher output is not improved productivity if defects, injuries, overtime or resource use increase disproportionately.

Explain Waste, value and continuous improvement and state one correct method, application or precaution.−

Waste is effort or resource use that does not create required customer value. Continuous improvement uses observation, data, experimentation and standardisation to improve the process safely.

Answer framework: Observe the work, distinguish value-added from necessary/non-value-added steps, identify a small testable improvement and record the outcome.

Important point: Never implement a productivity change without considering safety, quality and worker input.

Module 2 — Work Study and Work Measurement

Explain Method study and process charts and state one correct method, application or precaution.−

Method study records and critically examines how work is performed to eliminate unnecessary movement, delays and handling. Operation, flow, two-hand and worker-machine charts make hidden process steps visible.

Answer framework: Define scope, observe actual work ethically, use consistent symbols, ask purpose/place/sequence/person/means questions and propose a safer simpler method.

Important point: Do not measure or judge workers without transparency, consent through proper workplace process and attention to fair conditions.

Explain Motion economy and ergonomics and state one correct method, application or precaution.−

Motion-economy principles reduce unnecessary reach, search, force and fatigue. Ergonomic design considers human capability, posture, repetition, lighting, tools and work height.

Answer framework: Identify high-frequency motions, layout/reach issues and awkward postures; propose changes that reduce strain while preserving safety and quality.

Important point: Do not use an average worker assumption to ignore accessibility, fatigue or injury risk.

Explain Time study and standard time and state one correct method, application or precaution.−

Time study observes defined task elements, rates normal performance and adds appropriate allowances to establish a fair standard time. Work sampling estimates proportion of time spent in states through random observations.

Answer framework: Break task into observable elements, collect enough representative observations, document rating/allowance assumptions and report uncertainty.

Important point: A standard time is not a target to pressure workers; it must be reviewed when method, conditions or product changes.

Module 3 — Quality, Planning and Control

Explain Quality and variation and state one correct method, application or precaution.−

Quality means consistently meeting stated requirements. Variation may be common-cause (systemic) or special-cause (assignable); understanding the difference supports appropriate corrective action.

Answer framework: Define customer requirement and measurement method, collect data, plot/stratify it and use cause-and-effect or Pareto analysis before selecting action.

Important point: Do not adjust a stable process for every small fluctuation; over-adjustment can increase variation.

Explain Basic quality-improvement tools and state one correct method, application or precaution.−

Check sheets, Pareto charts, cause-and-effect diagrams, flowcharts, histograms and control charts support structured problem solving and evidence-based decisions.

Answer framework: Choose the tool that fits the question: frequency, distribution, sequence, causes or stability; state data source and sampling limitations.

Important point: Charts do not replace investigation—correlation, ranking or a visible pattern is not proof of root cause.

Explain Production planning and inventory and state one correct method, application or precaution.−

Planning coordinates demand, capacity, materials, routing, scheduling and inventory. Inventory buffers uncertainty but creates holding, obsolescence and cash costs; schedules must respect real constraints.

Answer framework: List demand, lead time, capacity, batch/lot assumptions and constraints; prepare a simple sequence and identify the bottleneck/risk.

Important point: Do not promise schedules without checking material availability, quality capacity, maintenance and human workload.

Module 4 — Facilities, Safety and Improvement Projects

Explain Facility layout and material flow and state one correct method, application or precaution.−

Facility layout positions resources to support flow, safety, supervision, flexibility and cost. Product, process, cellular and fixed-position layouts suit different volume/variety conditions.

Answer framework: Draw a spaghetti/flow diagram, quantify trips or distance where appropriate and compare layout alternatives against safety, throughput and flexibility criteria.

Important point: Do not reduce travel distance by creating unsafe crossings, congestion or inaccessible emergency routes.

Explain Safety and risk control and state one correct method, application or precaution.−

Industrial safety identifies hazards, assesses risk and applies controls using the hierarchy of elimination, substitution, engineering controls, administrative controls and PPE.

Answer framework: Describe task, hazard, exposure and consequence; choose the highest feasible control, assign responsibility and verify effectiveness.

Important point: PPE is important but is not the first or only control for a serious hazard.

Explain Improvement project communication and state one correct method, application or precaution.−

An improvement project needs a clear problem, baseline, target, evidence, tested change, implementation plan and follow-up measure. Stakeholder involvement improves feasibility and ownership.

Answer framework: Prepare a one-page A3-style report: background, current state/data, root causes, countermeasures, owner/timeline, result and next review.

Important point: Do not claim savings or quality gains without a defined baseline and a sustained post-change check.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Industrial Engineering and Productivity.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Work Study and Work Measurement.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.

5. State one key definition from Module 3: Quality, Planning and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Facilities, Safety and Improvement Projects.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Systems thinking, productivity, operations, resources, value, waste, costs, ergonomics and responsible improvement..
2. Develop a complete answer or practical plan for: Method study, process charts, motion economy, time study, work sampling, allowances and standard time..
3. Develop a complete answer or practical plan for: Quality concepts, variation, basic quality tools, process capability awareness, forecasting, scheduling, inventory and project control..
4. Develop a complete answer or practical plan for: Facility layout, material handling, line balancing, safety systems, risk awareness, project communication and improvement implementation..

LAST-MILE REVISION

Quick Revision

Module 1: Industrial Engineering and Productivity

Systems thinking, productivity, operations, resources, value, waste, costs, ergonomics and responsible improvement.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Work Study and Work Measurement

Method study, process charts, motion economy, time study, work sampling, allowances and standard time.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Quality, Planning and Control

Quality concepts, variation, basic quality tools, process capability awareness, forecasting, scheduling, inventory and project control.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Facilities, Safety and Improvement Projects

Facility layout, material handling, line balancing, safety systems, risk awareness, project communication and improvement implementation.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Industrial-engineering systems view	Industrial engineering studies the interaction of people, materials, machines, methods, information and environment.
Productivity and performance measures	Productivity relates output to inputs such as labour, material, energy or capital.
Waste, value and continuous improvement	Waste is effort or resource use that does not create required customer value.
Method study and process charts	Method study records and critically examines how work is performed to eliminate unnecessary movement, delays and handling.
Motion economy and ergonomics	Motion-economy principles reduce unnecessary reach, search, force and fatigue.
Time study and standard time	Time study observes defined task elements, rates normal performance and adds appropriate allowances to establish a fair standard time.
Quality and variation	Quality means consistently meeting stated requirements.
Basic quality-improvement tools	Check sheets, Pareto charts, cause-and-effect diagrams, flowcharts, histograms and control charts support structured problem solving and evidence-based decisions.
Production planning and inventory	Planning coordinates demand, capacity, materials, routing, scheduling and inventory.
Facility layout and material flow	Facility layout positions resources to support flow, safety, supervision, flexibility and cost.
Safety and risk control	Industrial safety identifies hazards, assesses risk and applies controls using the hierarchy of elimination, substitution, engineering controls, administrative controls and PPE.
Improvement project communication	An improvement project needs a clear problem, baseline, target, evidence, tested change, implementation plan and follow-up measure.

LEARNING RESOURCES

References

Recommended industrial-engineering references

1. M. P. Groover, Work Systems and the Methods, Measurement and Management of Work.
2. Mikell P. Groover, Automation, Production Systems, and Computer-Integrated Manufacturing.
3. R. C. Gupta, Industrial Engineering and Management.

Approved public industrial-engineering sources

- https://catalog.udayton.edu/allcourses/iet/iet.pdf
- https://ise.rpi.edu/programs/isye-courses

These sources support the study handbook while official Course 4023 content is unavailable; they do not replace official syllabus requirements.